



# Site selection decision framework for biomass pyrolysis project based on a mixed method under probabilistic linguistic environment and low carbon perspective: A case study in China

Shi-guan Yang, Jia-le Zhou, Zhuang Hu, Xin-yue Zhou, Qi Cai, Jin-heng Xie, Yang-wen Wu<sup>\*</sup>, Qiang Lu<sup>\*\*</sup>

National Engineering Research Center of New Energy Power Generation, North China Electric Power University, Beijing, 102206, PR China

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## ABSTRACT

Due to the shortage of fossil energy, pyrolysis has become a promising way for the thermal utilization of biomass resources, so as to partly replace fossil energy as well as achieve efficient resource usage. The site selection of the biomass pyrolysis (BP) project is an important task in the preliminary planning stage of the project. Moreover, since low-carbon development has become a common global goal at present, this paper proposes a decision-making framework from the low-carbon point of view for BP project site selection. Firstly, a complete site selection decision-making criteria system embracing the novel concept of low-carbon development is established for the first time, and the subjective and objective weighting method is integrated into the comprehensive integrated weighting method based on game theory. More importantly, a brand-new distance formula for the probabilistic linguistic term set is defined to measure the deviation between information, which uses expectation, variance and scale function to make the measurement more reasonable. Based on this, an improved TODIM (an acronym in Portuguese of interactive and multi-criteria decision making) method is proposed to more accurately determine the best option. Finally, taking Guangdong Province as an example, the best choice for site selection is obtained. This work can provide insightful theoretical basis for the site selection of BP projects.

## 1. Introduction

With the development of economy, population, urbanization and industrialization, the utilization of biomass resources is becoming an increasing concern in a wide range due to its strong resource conversion ability and zero-carbon-emission feature [1,2]. The thermal treatment technologies of biomass have gained widespread interests, among which the pyrolysis is far superior in terms of resource recovery and pollution prevention [3]. Pyrolysis of biomass is a thermal decomposition process in the absence of oxygen or with limited oxygen, to generate various products including bio-oil, bio-char and syngas, it provides opportunities for diversified biomass energy utilization [5]. Zhang et al. [6] analyzed the advantages of pyrolysis over incineration from the perspective of secondary pollution, and concluded that unlike incineration, the pyrolysis process was controlled by an anaerobic environment, which greatly reduced the content of polluting gases such as SO<sub>x</sub>, NO<sub>x</sub>, and HCl. Khan and Kabir [7] evaluated the economics of various

biomass waste disposal technologies, and found that pyrolysis technology was the most economical one since it could rationally apply biomass waste to form renewable resources and high-valued by-products, hence receiving extensive attention [4]. In summary, as the fourth largest energy consumption in the world, biomass energy plays an important role in the energy structure. The pyrolysis technology can convert low grade biomass into high grade products at low costs, which has attracted wide attention from various aspects.

For the development of biomass pyrolysis (BP), site selection is a key issue for the entire project. There are many factors that cannot be ignored in a BP project, which directly affects the subsequent economic benefits. Hence, enough attention must be paid to maximize benefits. Site selection for BP project is a multi-criteria decision-making (MCDM) process, which is generally constrained by multiple factors [8,9], yet little attention has been concentrated on the rationality of the location for a BP project. Therefore, this paper constructs a decision-making study on BP project site selection. Fortunately, the research on waste incineration project has achieved fruitful results, which can provide

<sup>\*</sup> Corresponding author.

<sup>\*\*</sup> Corresponding author.

E-mail addresses: [zorowuyangwen@163.com](mailto:zorowuyangwen@163.com) (Y.-w. Wu), [qianglu@mail.ustc.edu.cn](mailto:qianglu@mail.ustc.edu.cn), [qlu@ncepu.edu.cn](mailto:qlu@ncepu.edu.cn) (Q. Lu).

### Nomenclature

|                     |                               |
|---------------------|-------------------------------|
| $L(p)$              | probability of the PLTS       |
| $\#L(p)$            | the number of items           |
| $e(L(p))$           | expectation                   |
| $\sigma(L(p))$      | variance                      |
| $g(s_i)$            | PLTSs scale function          |
| $\omega_{jr}$       | relative criteria weights     |
| $\varphi_j$         | dominance of indicator        |
| $\theta$            | loss aversion coefficient     |
| $d(x_{pj}, x_{qj})$ | distance between two criteria |
| $T(A_p)$            | advantage of the scheme       |
| $S(A_p)$            | scheme sort value             |
| $CI$                | consistency indicator         |
| $RI$                | randomness indicator          |
| $\tau$              | number of elements            |

adequate references for BP project. During the MCDM process, the study of fuzzy set, decision method and weight determination method is extremely crucial for the site selection, and various researches have been conducted considering these factors. For example, Wu et al. [10] applied hesitant fuzzy linguistic term set to describe the evaluation information, combined analytic hierarchy process (AHP) and entropy weight method to determine the criteria weight, the fuzzy synthetic evaluation (FSE) method was used to analyze the comprehensive benefits of power plants from the perspective of garbage classification. Additionally, they also used interval binary semantics to describe the evaluation information, and then AHP and VIKOR (an acronym in Slovenski of interactive and multi-criteria decision making) methods were combined to establish a framework for waste-to-energy plant siting [11]. Table 1 summarizes the fuzzy sets, decision-making methods and weight-determining methods that have been used in waste incineration projects, which can provide a reference for the selection of these factors in this paper.

Compared with waste incineration, BP can not only reduce the environmental pollution but also improve the resource utilization. Biomass resources is an important renewable resource and a carbon

neutrality carrier, which plays an irreplaceable role in promoting carbon dioxide emission reduction. Moreover, building a clean and low-carbon energy system is an important measure for energy development in the new era. In this work, a MCDM framework for BP site selection from a low-carbon perspective is proposed. The implications of this work can be summarized in the following three aspects. First of all, the current studies on BP mainly focus on the reaction mechanism, reaction device, life cycle assessment, etc. Limited studies regarding on the site selection of the BP project have been reported in the literature survey. This paper provides scientific ideas and theories for the risk management of the project concentrating on the technical features of the BP projects. Secondly, low-carbon development has become a common global goal, and research on low-carbon is booming. For the first time, this paper combines low-carbon and BP project site selection, and considers the impact of low-carbon factors on site selection. Last but not least, the application of the existing distance formulas during MCDM process has encountered some problems such as distortion of original information and the different levels of differences, which leads to inaccurate calculation results. In order to improve the measurement accuracy, this paper introduces the expectation and variance into the distance formula, and uses the probabilistic language scaling function to define a novel distance measure, which improves the resolution accuracy of a reasonable measure.

The rest of the study is organized as follows. Section 2 summarizes the main research points through literature survey. Section 3 establishes the criteria system of the BP project from six aspects: policy, economy, society, technology, nature and low carbon. Section 4 introduces concepts related to the Probabilistic Linguistic Terms Sets (PLTSs) and puts forward a new PLTSs distance formula. In Section 5, based on the improved TODIM method, a framework is established for the BP site decision-making. Section 6 takes Guangdong Province as an example to examine the effectiveness of our approach. Section 7 draws conclusions.

## 2. Literature review

This section summarizes the three research points of the ambiguity in describing information and the choice of the decision method through literature survey, and makes a detailed analysis.

### 2.1. The ambiguity in describing information

The difficulty of decision-making for BP project mainly stems from the uncertainty of decision-makers' different preferences on the specific criteria. Uncertainty is manifested in two aspects. First, it is an ex-ante estimates of future value, so the value obtained is uncertain. Second, some subjective assessments come from the experience of decision-makers, but decision-makers cannot accurately express their opinions when describing complex objects, and there is always ambiguity in the way of thinking. Therefore, fuzzy set theory is introduced to describe decision information.

The classical fuzzy sets were first proposed by Zadeh [19], and fuzzy languages are constantly being supplemented and extended, such as two-dimensional semantics [12], intuitionistic fuzzy sets [20,21], interval triangular fuzzy numbers [22], hesitant fuzzy linguistic term sets [23], PLTSs [24], etc. Among these languages, PLTSs are widely used to deal with information ambiguity [29]. Liu and Li [25] established a ratio analysis of multi-objective optimization under prospect theory based on PLTSs. Lin et al. [26] used PLTSs to describe qualitative information in educational organization selection. Yu et al. [27] proposed a new likelihood formula for the first time by exploiting the properties of PLTSs. Liao et al. [28] verified the superiority of the new PLTS operator in the context of probabilistic linguistic. Krishankumar et al. [30] established a group decision framework for renewable energy selection based on interval-valued PLTS. The PLTS is a flexible structure that allows partial ignorance of occurring probabilities [31]. In conclusion, PLTSs can provide probabilistic information for linguistic terms and well preserve

**Table 1**  
Site selection decision-making and risk assessment of waste incineration projects.

| Fuzzy set                               | Decision method                                                               | Determination method           | Ref. |
|-----------------------------------------|-------------------------------------------------------------------------------|--------------------------------|------|
| Hesitant fuzzy linguistic terminology   | FSE                                                                           | AHP and entropy weight method  | [10] |
| Interval binary semantics               | VIKOR                                                                         | AHP                            | [11] |
| Triangular intuitionistic fuzzy numbers | Preference ranking organization method for enrichment evaluations (PROMETHEE) | Entropy weight                 | [12] |
| Triangular fuzzy number                 | Decision-making trial and evaluation laboratory (DEMATEL)                     | Analytic network process (ANP) | [13] |
| Triangular fuzzy number                 | Technique for order of preference by similarity to ideal solution (TOPSIS)    | AHP                            | [14] |
| Cloud model                             | Linear weighting                                                              | AHP                            | [15] |
| Hesitant fuzzy linguistic terminology   | TODIM                                                                         | ANP and entropy weight method  | [16] |
| Hesitant fuzzy linguistic terminology   | TOPSIS                                                                        | ANP and least squares          | [17] |
| Triangular fuzzy number                 | FSE                                                                           | AHP                            | [18] |

the initial information. Therefore, this paper uses PLTSs to describe the ambiguity of the evaluation process.

## 2.2. The choice of the decision method

The choice of the decision method is also an important step in MCDM. Three types of ranking methods are widely used at present, which is based on expected utility theory, level above relationship, and prospect theory (see Table 2).

As can be seen from the above table, the general method does not premeditate the influence of the psychological factors of decision-makers. In fact, the TODIM method which considers the psychological of decision makers, is more popular since it can capture the risk aversion of investors. Wu et al. [37] included the criterion weight determined using the entropy method in the site selection of offshore wind-solar-sea water pumped storage power stations in TODIM, and they also combined TODIM and PROMETHEE-II to select the best waste incineration project site [39]. Zhou et al. [38] also used the TODIM method to evaluate the integrated energy system of gas-wind-light-hydrogen. In Krishankumar's study, renewable energy source selections were prioritized by applying the TODIM method to q-rung orthopair fuzzy sets [41]. The decision-makers' psychological factors usually affect the decision-making results. Therefore, the TODIM method is more suitable for the sorting of the BP project.

## 2.3. Findings and contribution of this work

It can be found that the evaluation models still have some space for improvement, which can be summarized as follow.

- (1) Site selection is a complex issue, influenced by many factors, and the analysis of the influencing factors is not comprehensive enough. Most of the above literature uses the traditional politics, economics, sociology, technology (PEST) analysis method to construct a site selection decision-making criteria system, but does not consider the problem from a low-carbon perspective. Hence, the established criteria system is not comprehensive and perfect, especially when carbon emission reduction has become a global consensus.
- (2) The above literatures mostly used triangular intuitionistic fuzzy numbers, hesitant fuzzy numbers, and other forms to describe qualitative indicators. However, the evaluation of qualitative indicators depends on the perception of experts, who tends to use

linguistic terms to express their opinions. Furthermore, BP project site selection is a forward-looking affair, and there are many uncertainties.

- (3) Classical MCDM methods usually think that decision-makers are completely rational. However, decision-makers are usually faced with various risk states and uncertainties in the MCDM process, and the behavior of decision-makers is only bounded rationality.
- (4) The existing widely used weight assignment methods such as AHP, step-wise weight assessment ratio analysis (SWARA) do not consider the interaction among indicators. In fact, the sources of factors influencing BP site selection are complex and diverse, and there are often interdependent and causal relationships among them. To determine the weight only through the subjective weighting method will make the results less persuasive.

The ignorance of these problems will decrease the rationality of the results. In response to the above problems, this work constructs a framework for BP project, the contributions and innovation are summarized from four aspects.

- (1) Besides the traditional perspectives of environmental impact, technical level and public attitude, this paper innovatively takes the low-carbon factors as part of the consideration and also embraces the impact factors of politics, economy, society, and technology. It expands the site selection angle and enriches the application of low-carbon theory in the energy and environment field.
- (2) For the ambiguity of the actual decision environment, this work uses PLTSs to evaluate qualitative indicators, which not only allows the experts to express preferences, but also provides corresponding probability, to better preserve the original evaluation information.
- (3) To solve the problem of insufficient discrimination in the existing distance formula, a new distance formula is defined and applied in the TODIM method. This method is more in line with the real environment while considering the loss aversion behavior of decision-makers.
- (4) The ANP method is simple in operation and logically decomposes complex decision-making problems into small and related sub-problems to form a multi-level hierarchy. The subjective weighting method of ANP is employed to take into account the mutual relationship among the indicators, while the entropy method can reduce the deviation caused by excessive subjective factors. Therefore, this paper combines the ANP and the entropy method to determine the indicator weight.

**Table 2**

Comparative analysis of commonly used decision methods.

| Theory                   | Method                | Difference                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|--------------------------|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Expected utility theory  | Linear Weighting [32] | The global evaluation value of each item is directly obtained through the set counter, which is prone to information loss and distortion during the conversion process. The final ranking value of each item can be obtained through calculation, and the reason why each item satisfies the condition or does not meet the condition cannot be specifically displayed, which is not conducive to improving the items in the lower ranking. |
|                          | TOPSIS [33]           |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|                          | VIKOR [34]            |                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Level above relationship | ELECTRE [35]          | The method solves the compensation problem between criteria to a certain extent, but the calculation process is more complicated, and more parameters need to be considered.                                                                                                                                                                                                                                                                |
|                          | PROMETHEE [36]        | The indicators do not need to be dimensionless and standardized.                                                                                                                                                                                                                                                                                                                                                                            |
| Prospect theory          | TODIM [37]            | The method takes into account the decision maker's reference dependence and loss aversion.                                                                                                                                                                                                                                                                                                                                                  |

## 3. A criteria system for BP site selection

To meet the current environmental needs, a criteria system for BP is established based on a low-carbon perspective in this section.

Through a large number of literature surveys in the first stage, frequency statistics of indicators appearing in the previous literature (Table S1 in Supplementary Material) are conducted, and general influencing factors are summarized, namely political, economic, social, technological, and natural factors. Moreover, since the proposal of carbon peaking and carbon neutrality goals, low-carbon influencing factors cannot be ignored. The criteria system is built (see Fig. 1).

### 3.1. Policy factors C1

Land policy C11. It means that each local government firstly controls the total scale of construction land within a certain period according to the overall land-use plan, and manages it through the annual land-use plan.

Laws and regulations C12. Laws and regulations are important macro-influencing factors affecting project site selection, they provide a

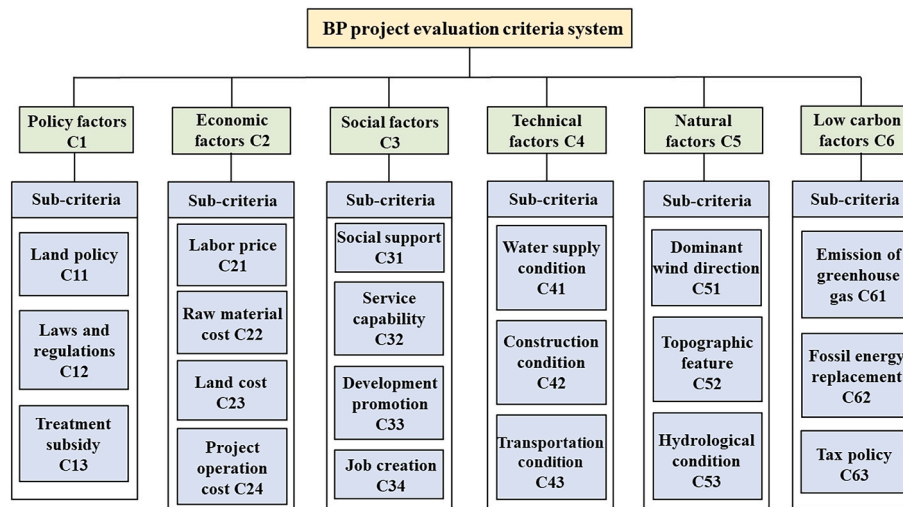


Fig. 1. Criteria system of site selection of BP.

code of conduct for project site selection. Before selecting a project site, it is vital to deeply understand the laws and regulations to ensure that the selected area allows the construction and development of BP projects.

**Treatment subsidy C13.** Only with the support of relevant policies can the market of the biomass waste treatment industry be guaranteed, the industry can develop rapidly, and the cash flow pressure of corporate investment can be effectively reduced.

### 3.2. Economic factors C2

**Labor price C21.** The operation of the BP project requires labors, and the collection and transportation of biomass resources also require a lot of labors.

**Raw material cost C22.** As the energy source of the BP, the supply of available biomass resources directly affects the economics of pyrolysis. Therefore, BP projects should be preferentially installed in areas with a large amount of biomass resources for a long time.

**Land cost C23.** The area of the alternative plant site is closely related to the construction scale of the pyrolysis project. Therefore, when considering the location, it is necessary to control the land area and price under the premise of complying with laws, regulations and local planning, so as to optimize the economic benefits of the BP project.

**Project operating cost C24.** It mainly includes the initial total investment before the completion of the project and the expenses required for project operation and maintenance. It is one of the goals of the construction of BP projects to complete the optimal plant site planning with the least economic investment.

### 3.3. Social factors C3

**Social support C31.** It refers to the attitude of the local government and residents towards the BP project. BP project brings a series of negative effects such as air and noise pollution, which influence the lives of residents. Therefore, in order to ensure the smooth progress of the project, the local government and residents should be communicated in a timely manner during the site selection process.

**Service capability C32.** It is used to evaluate the service situation of the project to the local area and the necessary degree of construction. The stronger the service capability and the larger the service radius, the greater the social benefit of the project, which promotes and guarantees the long-term operation of the project to a certain extent.

**Development promotion C33.** The BP project uses biomass to provide products for society, and requires the participation of multiple suppliers

in the project implementation process. Therefore, the construction of the BP project promotes the growth of social gross domestic product and the development of the local economy.

**Job creation C34.** This indicator mainly involves the large demand for labor in various fields of biomass treatment, project planning, equipment layout, equipment maintenance, thus creating good social benefits in employment opportunities.

### 3.4. Technical factors C4

**Water supply condition C41.** Water resources have a crucial impact on the operation performance and energy consumption. Insufficient water supply not only leads to project delays, but also has a huge impact on safety and the economy. Therefore, when selecting the site, the water supply network of the alternative site should be inspected to ensure a good water supply.

**Construction condition C42.** In addition to the land used for the factory area, there should also be necessary construction sites near the factory area to create convenient conditions for speeding up the construction.

**Transportation condition C43.** When building an BP project, there is a large demand for transportation, relevant equipment and construction materials must be transported over long distances. The convenient transportation can further save the initial investment and the costs for operation and maintenance.

### 3.5. Natural factors C5

**Dominant wind direction C51.** The BP project produces polluting gas during operation. Attention should be paid to the influence of the local dominant wind direction on the speed and scope of pollution diffusion. Therefore, the BP project should be constructed in the downwind area with the dominant wind direction in the residential area.

**Topographic feature C52.** The stability of the BP project construction is evaluated by using topographical features, measured according to the results of on-site investigations and historical data, and described in the form of grades.

**Hydrological condition C53.** The hydrogeological condition directly affects the civil work of the BP project. On the other hand, loose soil brings difficulties to the project construction. Therefore, during the project site selection, it is necessary to evaluate the hydrological environment of each candidate site area to ensure that the requirements for construction and operation are met.



### 3.6. Low carbon factors C6

Emission of greenhouse gas C61. Compared with fossil energy, biomass is not only an important renewable resource, but also a carbon-neutral carrier, and pyrolysis technology can greatly reduce greenhouse gas emissions.

Fossil energy replacement C62. The BP project replaces the use of fossil energy such as coal, oil, and natural gas by using biomass, a zero-carbon-emission resource. On the one hand, the depletion of non-renewable energy is alleviated; on the other hand, the unnecessary waste of resources in biomass is reduced, and the degree of energy utilization is greatly improved.

Tax policy C63. Considering the site selection of BP projects under the carbon tax policy, increasing carbon tax increases the cost of enterprises. Under the “dual carbon” goal, it is only possible to achieve the goal of reducing emissions through a flexible carbon tax policy.

## 4. Methodology

This section analyzes the related theories of PLTSs. Moreover, a new distance formula for PLTSs is defined, which provides a theoretical basis for the follow-up research.

### 4.1. Probabilistic linguistic terminology

In real life, the evaluation languages given by different decision-makers for the same criteria are different. PLTSs take this into consideration and add the probability of their occurrence to the original linguistic terms to make the decision more reliable and accurate.

**Definition 1.** [40]  $S = \{s_i | 0, 1, \dots, \tau\}$  is set as a linguistic term set, then Pang et al. [24] defined a PLTS as:

$$L(p) = \left\{ L_k(p) | L_k \in S, p_k \geq 0, k = 1, 2, \dots, \#L(p), \sum_{k=1}^{\#L(p)} p^k \leq 1 \right\} \quad (1)$$

Where  $L_k(p)$  denotes the probability  $p_k$  of the linguistic term set.  $\#L(p)$  indicates the number of items contained in the linguistic term set.

**Definition 2.** For a given set  $L(p)$ , normalizes it so that the probabilities sum to 1. The normalized PLTS is obtained as  $L(\tilde{p})$ :

$$L(\tilde{p}) = \{L^{(k)}(\tilde{p}^k) | k = 1, 2, \dots, \#L(p)\} \quad (2)$$

Where  $(\tilde{p}^k) = p_k / \sum_{k=1}^{\#L(p)} p^k$ . For probabilistic sums of PLTSs less than 1, it can be written as  $\sum_{k=1}^{\#L(p)} p^k \leq 1$ .

**Definition 3.** Given that the  $L(p)$  be a PLTS with incomplete information, and its normalized form  $L(\tilde{p})$  is:

$$L(\tilde{p}) = \{L^{(k)}(\tilde{p}^k) | k = 1, 2, \dots, \#L(\tilde{p})\} \quad (3)$$

For the probability value equals to 1, it is a standard type, as shown in formula (3).

**Definition 4.** [42]. Defined the expectation as:

$$e(L(p)) = \sum_{k=1}^{\#L(p)} g(L(p)) \times p_k \quad (4)$$

**Definition 5.** Defined the variance as:

$$\sigma(L(p)) = \sum_{k=1}^{\#L(p)} (g(L(p)) - e(L(p)))^2 \times p_k \quad (5)$$

**Definition 6.** If  $S = \{s_i | 0, 1, \dots, \tau\}$  is a set, and  $s_i$  can be represented by the corresponding scale function:

$$g(s_i) = \begin{cases} \frac{a^\tau - a^{\tau-i}}{2a^\tau - 2}, i = 0, 1, \dots, \tau \\ \frac{a^\tau + a^{\tau-i} - 2}{2a^\tau - 2}, i = \tau + 1, \tau + 2, \dots, 2\tau \end{cases} \quad (6)$$

Among them, the usual value  $a$  is 1.37 [43].

### 4.2. Distance formulas for probabilistic linguistic term sets

#### (1) Existing distance measurement

Pang et al. [24] defined the degree of deviation of two PLTSs, the Euclidean distance between them is:

$$d(L_1(p), L_2(p)) = \sqrt{\sum_{k=1}^{\#L(p)} (p_{1k}r_{1k} - p_{2k}r_{2k})^2 / \#L_1(p)} \quad (7)$$

Where  $r_{1k}$  and  $r_{2k}$  are the subscripts of two PLTSs, and  $\#L_1(p)$  is the quantity contained in the PLTSs.

Lin and Xu [44] proposed the Hamming distance of probabilistic linguistic as:

$$d(L_1(p), L_2(p)) = \sum_{k=1}^{\#L(p)} |p_{1k}r_{1k} - p_{2k}r_{2k}| / \#L_1(p) \quad (8)$$

It is worth noting that both formulas (7) and (8) use the form of probability value multiplied by linguistic subscript value to perform corresponding subtraction of elements in the two PLTSs, but this form lacks rationality. When the two PLTSs contain different linguistic subscripts and probability values, and the product of the two is the same, the use of formulas (7) and (8) cannot effectively measure the deviation of the two PLTSs (see Example 1). In addition, the above Euclidean distance and Hamming distance both assume that both PLTSs contain the same number of elements. If they are different, element complementing processing is required, which may lead to inaccurate measurement results (see Example 1).

**Example 1.** Suppose  $S$  is a PLTS,  $L_1(p) = \{s_2(0.25), s_3(0.5), s_4(0.25)\}$  and  $L_2(p) = \{s_3(1)\}$  are two PLTSs defined on  $S$ . Before calculation, firstly add the element in  $L_2(p)$  as  $L_2(p) = \{s_3(0), s_3(0), s_3(1)\}$ , use formula (7) to calculate that  $d(L_1(p), L_2(p)) = 0.85$ , use formula (8) to calculate that is 1.33, and the deviation between them is large. Whereas, they contain the same expected value, only the variance is different, and the accuracy of the distance between the two PLTS calculated by formulas (7) and (8) needs to be improved.

To solve the above problem, Zhang et al. [42] defined the distance formula as:

$$d\left(L_1\left(p\right), L_2\left(p\right)\right)=\sum_{k=1}^{\# L(p)} p\left(r_{1 k}, r_{2 k}\right) p\left(r_{1 k}, r_{2 k}\right) \quad (9)$$

Where  $d(r_{1k}, r_{2k}) = \frac{r_{1k} - r_{2k}}{\tau}$ ,  $p(r_{1k}, r_{2k}) = p_{1k}, p_{2k}$ ,  $\tau$  is the number of elements. The difference between the subscripts of two adjacent linguistic terms is 1, and by substituting  $d(r_{1k}, r_{1k})$  into formula (8), it can be found that the gap between any two adjacent linguistic terms is equal, equal to  $\frac{p(r_{1k}, r_{2k})}{\tau}$ . In some practical problems, however, the degree of difference between levels is not the same. For example, for a variable that obeys a normal distribution, the difference in expected value between adjacent levels determines the position of the normal curve, thereby influencing the magnitude of the normal curve distribution. In addition, when two PLTSs contain the same linguistic value but different probability values, formula (8) cannot effectively measure the deviation of the two PLTSs (see Example 2).

**Example 2.** Let  $S = \{s_i | 0, 1, 2, 3, 4\}$  be a PLTS,  $L_1(p) = \{s_2(0.3), s_3(0.7)\}$  and  $L_2(p) = \{s_2(0.5), s_3(0.5)\}$  based on  $S$ . Calculated  $d(L_1(p), L_2(p)) = 0$  by formula (8). Obviously, both PLTSs contain  $s_2, s_3$  but their corresponding probabilities are not the same, so it is unreasonable.

#### (2) Improved distance measure

It can be seen from the above analysis that the existing distance formula has the problem of calculation inaccuracy. To improve the measurement accuracy, this paper introduces the expectation and variance into the distance formula and uses the probabilistic linguistic scaling function [43] to define the distance measure. At the same time, in actual decision-making, people always believe that the larger the  $e(L(p))$  better, and the smaller the  $\sigma(L(p))$  better. This paper improves the proposed distance formula and proposes a new probabilistic linguistic distance formula. The distance formula can solve the problem of deviation among the grades in the literature [42]. In addition, it can also solve the problem of adding new linguistic terms when the lengths are different in the literature [44].

**Definition 7.** If  $S = \{s_i | 0, 1, \dots, \tau\}$  is a linguistic term set, the distance between  $L_1(p)$  and  $L_2(p)$  is:

$$d(L_1(p), L_2(p)) = \left| \frac{e(L_1(p))}{\sigma(L_1(p)) + 1} - \frac{e(L_2(p))}{\sigma(L_2(p)) + 1} \right| \quad (10)$$

$$e(L(p)) = \sum_{k=1}^{\#L(p)} g(L(p)) \times p_k$$

$$\sigma(L(p)) = \sum_{k=1}^{\#L(p)} (g(L(p)) - e(L(p)))^2 \times p_k$$

$$g(L(p)) = \begin{cases} \frac{a^\tau - a^{\tau-i}}{2a^\tau - 2}, i = 0, 1, \dots, \tau \\ \frac{a^\tau + a^{\tau-i} - 2}{2a^\tau - 2}, i = \tau + 1, \dots, 2\tau \end{cases}$$

Where  $e(L(p))$  is the expectations and  $\sigma(L(p))$  is the variances.  $g(L(p))$  is the scale function of the probabilistic linguistic term.

The relationship between the distance formula shown in formula (10) and the existing distance formula can be summarized as follows. Similarities:

- 1) The new distance formula and the existing distance formula both consider the linguistic terminology, which is closer to the actual decision-making process.
- 2) The new distance formula and the existing distance formula both take into account the probability information of the linguistic term set, according to which PLTSs are converted to real numbers.

Differences:

- 1) In this paper, the proposed distance formula introduces the expected value and the variance value into the distance formula, which can improve the resolution accuracy. This distance formula solves two types of problems in the existing distance formulas, that is, when the number of elements contained in the PLTSs is unequal, it is necessary to add several PLTSs with a probability of 0 to make it convenient to calculate and consider the lack of differentiation caused by the grade deviation.
- 2) Compared with the existing distance formulas, the new distance formula introduces the probability linguistic scale function to make the measure more reasonable.

#### 4.3. Improved TODIM method

The classic TODIM method can reflect the psychological behavior of decision-makers, and its core is to calculate the dominance degree between each scheme and other schemes. This paper improves it as follows:

**Step1.** Calculate the relative criteria weights. The relative properties of the indicator  $b_i$  relative to the indicator  $b_r$  are defined as:

$$\omega_{jr} = \frac{\omega_j}{\omega_r} \quad (11)$$

Where  $\omega_r = \max\{\omega_j | j = 1, 2, \dots, l\}$ ,  $\omega_j$  is the objective weight.

**Step2.** Calculate the dominance of each indicator. This paper uses improved distance measures to determine deviations between scenarios. Therefore, the advantages of the factory location  $A_p$  for  $A_q$  in the indicator  $b_j$  are:

$$\delta(A_p, A_q) = \sum_{j=1}^m \varphi_j(A_p, A_q) \quad (12)$$

$$\sum_{j=1}^m \varphi_j(A_p, A_q) = \begin{cases} \sqrt{\frac{d(x_{pj}, x_{qj}) \omega_{jr}}{\sum_{j=1}^m \omega_{jr}}}, & \text{if } x_{pj} - x_{qj} > 0 \\ 0, & \text{if } x_{pj} - x_{qj} = 0 \\ -\frac{1}{\theta} \sqrt{\frac{d(x_{pj}, x_{qj}) \sum_{j=1}^m \omega_{jr}}{\omega_{jr}}}, & \text{if } x_{pj} - x_{qj} < 0 \end{cases} \quad (13)$$

In formula (13), the parameter  $\theta$  is the loss aversion coefficient,  $d(x_{pj}, x_{qj})$  represents the distance between two alternative criteria values.

**Step3.** Calculate the dominance of the entire program. Based on the above results, it can be obtained that the overall dominance of the factory location  $A_p$  relative to  $A_q$  is:

$$T(A_p) = \sum_{q=1}^m \delta(A_p, A_q) \quad (14)$$

**Step4.** Calculate the scheme sort value. By normalizing the overall dominance  $T(A_p)$ , it is possible to obtain the scheme sort values as:

$$S(A_p) = \frac{T(A_p) - \min T(A_p)}{\max T(A_p) - \min T(A_p)} \quad (15)$$

Then the alternatives are sorted based on the size of the  $S(A_p)$  value.

#### 5. Site selection decision-making framework for BP

This section establishes a four-stage framework for BP project site selection, as shown in Fig. 2.

##### 5.1. Phase 1 - identify alternative locations

**Step1.** Form an expert committee. Experts from various fields such as pyrolysis, engineering economics, environment, and operations research with academic backgrounds are invited.

**Step 2.** Determine the alternative site for the BP project. Before proceeding with the project site selection, the decision-makers first determine the initial candidate site according to their own conditions, and exclude some alternative projects that are not supported by the government. Eventually the selection of alternative plant sites is completed.

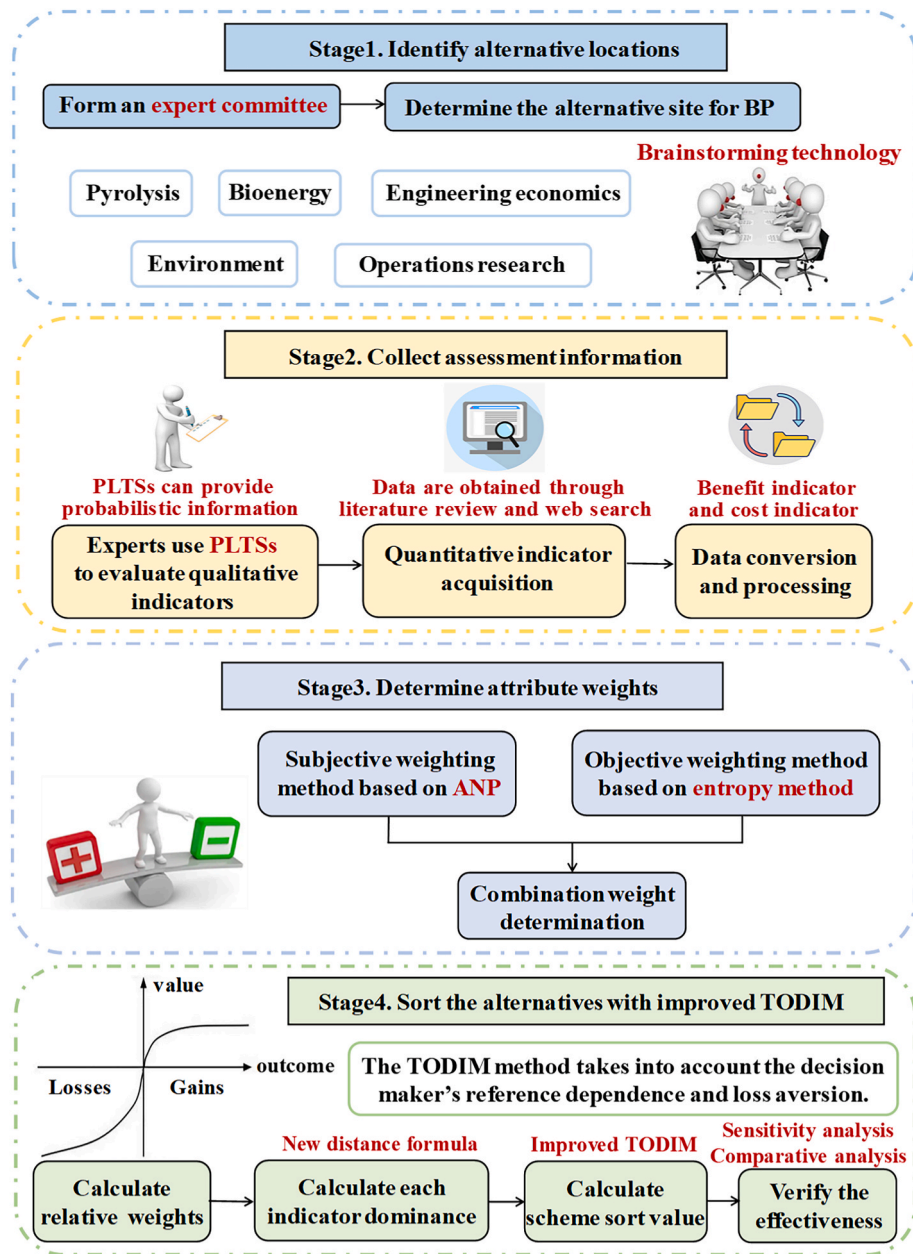


Fig. 2. Site selection decision-making framework for BP.

### 5.2. Phase 2 - collection of initial evaluation information from experts

**Step1.** Qualitative indicators need to be determined through expert scoring. The expert committee uses 5-granularity linguistic term sets to evaluate each indicator  $S = \{s_1, s_2, \dots, s_n\}$ , and delineates unified standards:

$$S = \{s_0 : \text{Poor}, s_1 : \text{Medium Poor}, s_2 : \text{Medium}, s_3 : \text{Medium Good}, s_4 : \text{Good}\}$$

**Step2.** Quantitative indicator acquisition. Quantitative indicator data are obtained through literature review and web search.

**Step3.** Data transformation and processing. Since the units of quantitative data are different, the data should be processed without dimension. Assuming that there are  $n$  schemes,  $m$  indicators, and  $x_{ij}$  represents the evaluation value of the  $j^{\text{th}}$  indicator in the  $i^{\text{th}}$  scheme, the data needs

to be normalized, and the formula is as follows:  
Benefit indicator:

$$y_{ij} = \frac{x_{ij} - \min x_{ij}}{\max x_{ij} - \min x_{ij}} \quad (16)$$

Cost indicator:

$$y_{ij} = \frac{\max x_{ij} - x_{ij}}{\max x_{ij} - \min x_{ij}} \quad (17)$$

Eventually, the evaluation value of each quantitative indicator is given by the expert committee. A complete decision matrix of the PLTS is obtained.

### 5.3. Phase 3 - determining criteria weights

Based on the respective advantages and disadvantages of the ANP and entropy weighting method, this work uses the optimal combination to determine the final weight.

**Table 3**

Level 1–9 scaling scale.

| Importance | Definition                                       |
|------------|--------------------------------------------------|
| 1          | $C_i$ and $C_j$ are equally important            |
| 3          | $C_i$ is slightly more important than $C_j$      |
| 5          | $C_i$ to $C_j$ is important                      |
| 7          | $C_i$ to $C_j$ is very important                 |
| 9          | $C_i$ to $C_j$ is extremely important            |
| 2,4,6,8    | The median value of the above adjacent judgments |

### 5.3.1. Weight determination based on ANP

**Step1.** Pairwise comparison of indicators. According to the 1–9 scale (Table 3), the indicators are compared and scored in pairs, and the importance of the indicators is repeatedly determined by the expert committee.

**Step2.** Consistency check. After the pairwise comparison of the indicators, a consistency check is required to test the reliability of the results. The formula is:

$$CR = \frac{CI}{RI} \quad (18)$$

Where  $CI$  is the consistency indicator, and  $RI$  is the randomness indicator. When  $CI < 0.1$ , the pairwise comparison matrix is consistent.

**Step3.** Calculate the criteria weight. After the consistency check, the decision super matrix of the ANP structure is generated, and the subjective weight can be obtained.

### 5.3.2. Weight determination based on entropy weight method

**Step1.** Calculate the information entropy of the  $j^{th}$  indicator  $e_j$ :

$$e_j = -\frac{1}{\ln n} \sum_{i=1}^m \left( \frac{y_{ij}}{\sum_{i=1}^m y_{ij}} \times \ln \left( \frac{y_{ij}}{\sum_{i=1}^m y_{ij}} \right) \right) \quad (19)$$

**Step2.** Calculate the objective weight of the  $j^{th}$  indicator  $\omega_j$ :

$$g_j = 1 - e_j \quad (20)$$

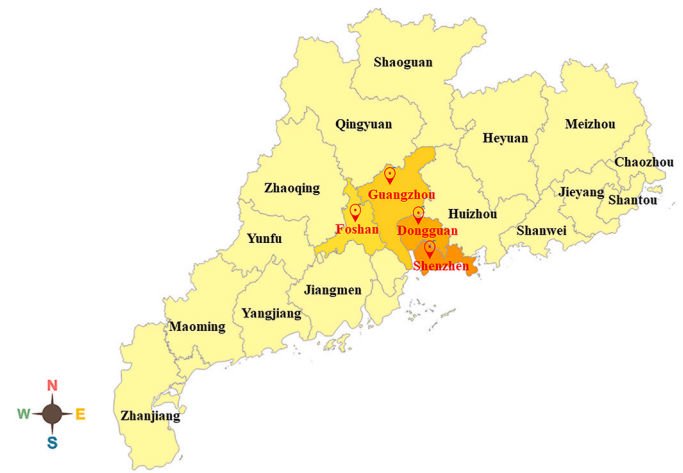
$$\omega_j = \frac{g_j}{\sum_{j=1}^n g_j} \quad (21)$$

### 5.3.3. Combination weight by the comprehensive integrated weighting method

After obtaining the weights, the optimization target model [45] is used for combined weighting:

$$\min \|\alpha_1 \omega_1 + \alpha_2 \omega_2 - \omega_j\|_2 (\alpha_j > 0, j = 1, 2) \quad (22)$$

**Step1.** Determine the optimal first derivative condition and the corresponding linear equation system:

**Fig. 3.** The specific geographic location of the alternative site.

$$\sum_{i=1}^2 \alpha_i \omega_j \omega_i^T = \omega_j \omega_i^T (j = 1, 2) \quad (23)$$

$$\begin{bmatrix} \omega_1 \omega_1^T & \omega_1 \omega_2^T \\ \omega_2 \omega_1^T & \omega_2 \omega_2^T \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \begin{bmatrix} \omega_1 \omega_1^T \\ \omega_2 \omega_2^T \end{bmatrix} \quad (24)$$

**Step2.** Calculate and normalize the parameters to get  $\alpha_1^*$  and  $\alpha_2^*$ . Then the weight of each indicator combination is:

$$\omega = \alpha_1^* \omega_1 + \alpha_2^* \omega_2 \quad (25)$$

Herein, it needs to meet the condition  $\alpha_1^* + \alpha_2^* = 1 (\alpha_1^*, \alpha_2^* > 0)$ .

### 5.4. Phase 4 - the improved TODIM method for scheme ranking

The improved TODIM method consists of four steps, which are detailed in 4.3.

## 6. A case study

### 6.1. Project information introduction

The Guangdong Provincial Government is preparing to build a new BP project. An enterprise obtains the consent of the Guangdong Provincial Government to determined alternative sites. Its geographic location and specific information are shown below (Fig. 3 and Table 4).

### 6.2. Collection and processing of site selection decision-making data

Based on the constructed criteria system, the first step is to set up an expert committee which consists experts from various fields including pyrolysis, bioenergy, engineering economics, environment and operations research, and then ask the committee to score each indicator of the BP project. The indicators are expressed in probabilistic linguistic terms (see Table S2).

$$A = \begin{matrix} A1 \\ A2 \\ A3 \\ A4 \end{matrix} \begin{pmatrix} 2.8 & 3 & 3 & 2.8 & 2.3 & 3 & 1.9 & 2.7 & 2.2 & 2.9 & 3 & 3 & 2.4 & 3 & 3.6 & 1.7 & 3.6 & 3 \\ 3.5 & 2.7 & 2.7 & 2 & 1.5 & 1.8 & 3.2 & 3.2 & 2 & 2 & 2 & 3 & 1.8 & 2 & 2 & 2 & 2 & 2.8 \\ 4 & 2.5 & 2.5 & 3 & 1.8 & 2.6 & 1.7 & 1.7 & 1.9 & 2.3 & 2.2 & 3 & 3 & 2.6 & 2.6 & 2.4 & 2.6 & 3 \\ 2.7 & 4 & 4 & 3.9 & 2.4 & 2.3 & 3 & 3 & 2.4 & 1.9 & 2.6 & 3.4 & 4 & 1.7 & 3 & 2.7 & 3 & 4 \end{pmatrix}$$



**Table 4**

Comparison of the relevant information of the four alternative sites (Source: China environmental sanitation technology network).

| Alternative site                                   | A1       | A2     | A3       | A4        |
|----------------------------------------------------|----------|--------|----------|-----------|
| Location                                           | Shenzhen | Foshan | Dongguan | Guangzhou |
| Population (10,000 people)                         | 1756     | 950    | 1047     | 1867      |
| Biomass production in the whole city (10,000 tons) | 712.4    | 436.2  | 449.0    | 808.8     |
| Occupied area (Square kilometers)                  | 1996.85  | 3875   | 2512     | 7434.4    |

Based on the basic information of the above four alternative sites, the company decides to use the BP project site selection decision-making framework from a low-carbon perspective constructed above to realize the process of site selection.

### 6.3. Criteria weight calculation

Through further analysis of the construction of the criteria system, the expert committee determines the correlation among the indicators and the subjective weight results calculated by the ANP method:

$$\omega_1 = (0.0287, 0.0547, 0.0369, 0.0702, 0.0475, 0.0533, 0.0702, 0.0585, 0.0329, 0.0358, 0.0599, 0.0541, 0.0460, 0.0549, 0.0446, 0.0614, 0.0473, 0.0369, 0.0287, 0.0776).$$

According to formula (19) to formula (21), the objective weight results calculated by the entropy weight method are:

$$\omega_2 = (0.0635, 0.0638, 0.0638, 0.0381, 0.0584, 0.0426, 0.0556, 0.0349, 0.0503, 0.0635, 0.0503, 0.0322, 0.0356, 0.0475, 0.0406, 0.0434, 0.0406, 0.0434, 0.0406, 0.0326).$$

According to the formula (25), and the parameters are  $\alpha_1 = 0.1786$  and  $\alpha_2 = 0.8271$ . The combined weight of each indicator is obtained by weighting. The final results are shown in Table 5.

### 6.4. Decision-making framework application calculation

The new distance formula for PLTSs is introduced into the improved TODIM method to rank the schemes. The distance between the indicator values of any two schemes is calculated by the formula (10) (see Table 6). Based on the data obtained, a comprehensive ranking value for the four candidate sites is then calculated according to the improved TODIM method steps as follows.

The dominance degree between the schemes under each indicator is calculated according to Tables 6 and 7, and formula (12). The calculation is shown in Table 8 (see Table S3 for other indicators).

Repeat the above steps, and then obtain the advantage matrix of the whole scheme according to formula (14) (see Table 9).

The ranking order of candidate sites can be obtained from Table 10 as follows:  $A4 > A1 > A3 > A2$ , and A4 is the best solution.

### 6.5. Sensitivity analysis of decision results

#### 6.5.1. Change the loss aversion factor

Since the  $\theta$  number of the loss avoidance coefficient reflects the different attitudes of decision-makers towards losses, the change of the  $\theta$  value will cause changes in the decision results of the BP project.

In this work, the loss avoidance coefficient  $\theta$  is set to 0.1, 0.3, 0.5, 0.7, 0.9, 1, 2, 3, 4, 5, 6, 7, 8, respectively. The sensitivity test is carried out on the constructed MCDM framework. To simulate the risk aversion attitude of different decision-makers in the actual decision-making environment, to test whether the change of the value of  $\theta$  will have a greater impact on the evaluation results, so as to test the stability of the model in different decision-making environments. The calculation results are shown in Table S4. With the change of the value of  $\theta$ , the sorting results are all  $A4 > A1 > A3 > A2$ , and A4 is always the best choice for enterprise location.

Through further observation, it can be found that with the change of the loss avoidance coefficient  $\theta$ , the ranking result does not change. The smaller the value of  $\theta$ , the larger the comprehensive ranking value of A1 and A3 items, while the values of A2 and A4 remain unchanged, that is, the larger the overall dominance gap between items. This also shows that as decision makers' degree of loss avoidance increases, there will be more stringent standards for a certain aspect of the project, and the disadvantages of each alternative project in this aspect will be more likely to appear. The gap between the alternatives becomes clear.

The alternatives in Fig. 4 can still maintain a stable ranking when the value is constantly changing  $\theta$ . A4 is the best location among all the alternatives, which verifies the BP project constructed in this paper from a low-carbon perspective.

The principle of constructing a comprehensive evaluation framework for multi-criteria decision-making is based on criteria weights. It is also essential to change the weights to perform sensitivity analysis on the evaluation indicators, and to verify the reliability of the framework by testing whether the fluctuation of the criteria weights will bias the decision results. In this paper, the weights of the 20 indicators fluctuate by 1% and 2% respectively, and the sensitivity test is carried out in a way of recalculating the ranking results. The impact of the fluctuation of each indicator on the results is shown in Fig. S1 to Fig. S6.

It can be seen intuitively that there is a slight gap between the dominance of some qualitative indicators (such as C33, and C34) and the dominance of other indicators, indicating that the satisfaction of the site A1 in terms of social factors is higher than that of the site A4. Some quantitative indicators (such as the ranking results of C61, and C62) are significantly different from other schemes, indicating that different sites place different emphasis on low carbon. However, deviations in individual metrics have little effect on the final result.

As the criteria weight fluctuates, the final results are almost  $A4 > A1 > A3 > A2$  (see Fig. S1 to Fig. S6). Through further observation, it can be found that the ranking results have not changed, and the project A4 is always the best site while verifying the stability of the framework.

### 6.6. Comparative analysis

We compare our method with TOPSIS [46] and the VIKOR [47]. For both methods, we adopt the initial data from Table S2 and Table 5. The TOPSIS method is used to calculate the relative proximity of the schemes, and the VIKOR method is used to calculate the compromise evaluation value of the schemes. The decision results and the corresponding ranking are compared as follows (see Table 11).

It can be known from Table 11 that the results of the three methods are slightly different, but the best solution calculated by these three methods is always A4, which can be proven the accuracy of this method. The final result obtained by the TOPSIS method and the VIKOR method

**Table 5**

Combined weights of criteria.

| Criteria | Subjective weight | Objective weight | Combined weight | Criteria | Subjective weight | Objective weight | Combined weight |
|----------|-------------------|------------------|-----------------|----------|-------------------|------------------|-----------------|
| C11      | 0.0287            | 0.0635           | 0.0577          | C34      | 0.0599            | 0.0503           | 0.0523          |
| C12      | 0.0547            | 0.0638           | 0.0626          | C41      | 0.0541            | 0.0322           | 0.0363          |
| C13      | 0.0369            | 0.0638           | 0.0594          | C42      | 0.0460            | 0.0840           | 0.0777          |
| C21      | 0.0702            | 0.0381           | 0.0441          | C43      | 0.0549            | 0.0564           | 0.0565          |
| C22      | 0.0475            | 0.0584           | 0.0568          | C51      | 0.0446            | 0.0356           | 0.0374          |
| C23      | 0.0533            | 0.0426           | 0.0448          | C52      | 0.0614            | 0.0475           | 0.0502          |
| C24      | 0.0702            | 0.0556           | 0.0586          | C53      | 0.0473            | 0.0406           | 0.0420          |
| C31      | 0.0585            | 0.0349           | 0.0393          | C61      | 0.0369            | 0.0434           | 0.0425          |
| C32      | 0.0329            | 0.0503           | 0.0475          | C62      | 0.0287            | 0.0406           | 0.0387          |
| C33      | 0.0358            | 0.0653           | 0.0604          | C63      | 0.0776            | 0.0326           | 0.0409          |

**Table 6**The distances between the alternative site indicators under each indicator  $C_{ij}^a$ .

|     | $d(A1, A2)$ | $d(A1, A3)$ | $d(A1, A4)$ | $d(A2, A3)$ | $d(A2, A4)$ | $d(A3, A4)$ |
|-----|-------------|-------------|-------------|-------------|-------------|-------------|
| C11 | 0.356       | 0.623       | 0.207       | 0.979       | 0.564       | 0.415       |
| C12 | 0.325       | 0.701       | 1.439       | 0.375       | 1.114       | 0.738       |
| C13 | 0.325       | 0.701       | 1.439       | 0.375       | 1.114       | 0.738       |
| C21 | 1.304       | 0.029       | 1.380       | 1.332       | 0.077       | 1.409       |
| C22 | 0.235       | 0.088       | 0.111       | 0.323       | 0.124       | 0.199       |
| C23 | 0.0786      | 0.530       | 0.565       | 0.609       | 0.644       | 0.035       |
| C24 | 0.313       | 0.313       | 0           | 0           | 0.313       | 0.313       |
| C31 | 0.600       | 0.354       | 0.038       | 0.246       | 0.638       | 0.392       |
| C32 | 0.021       | 0.091       | 0.137       | 0.07        | 0.117       | 0.047       |
| C33 | 0.479       | 0.638       | 0.359       | 0.158       | 0.120       | 0.279       |
| C34 | 0.156       | 0.425       | 0.515       | 0.268       | 0.359       | 0.090       |
| C41 | 0           | 0           | 0.225       | 0           | 0.225       | 0.225       |
| C42 | 0.093       | 0.163       | 0           | 0.256       | 0.093       | 0.163       |
| C43 | 0.489       | 0.163       | 0.640       | 0.326       | 0.152       | 0.477       |
| C51 | 1.527       | 0.325       | 1.114       | 1.852       | 0.413       | 1.439       |
| C52 | 0.506       | 0.564       | 0.074       | 0.059       | 0.431       | 0.490       |
| C53 | 0           | 0.985       | 0.985       | 0.985       | 0.985       | 0           |
| C61 | 0.355       | 0.355       | 0           | 0           | 0.355       | 0.355       |
| C62 | 0           | 0.985       | 0.985       | 0.985       | 0.985       | 0           |
| C63 | 0.318       | 0           | 1.126       | 0.318       | 0.807       | 1.126       |

<sup>a</sup> Among them, i is the primary indicator and j is the sub-indicator.

(1) Calculate the relative criteria weights, as shown in Table 7.

is the same, while the second and third places are reversed compared with the method of this paper, which indicates that the decision results are more favorable when the psychological factors of the decision-makers are taken into account. It is biased towards A1. At this time, A1 is the second-best site, which is more in line with the real environment.

Compared with the TOPSIS and VIKOR methods, our method has a

**Table 7**

Relative criteria weights.

(2) Calculate the dominance of each indicator.

|     | $\omega_{jr}$ |     | $\omega_{jr}$ |     | $\omega_{jr}$ |     | $\omega_{jr}$ |
|-----|---------------|-----|---------------|-----|---------------|-----|---------------|
| C11 | 0.755         | C23 | 0.507         | C34 | 0.599         | C52 | 0.878         |
| C12 | 0.759         | C24 | 0.662         | C41 | 0.383         | C53 | 0.483         |
| C13 | 0.759         | C31 | 0.415         | C42 | 1             | C61 | 0.517         |
| C21 | 0.454         | C32 | 0.599         | C43 | 0.671         | C62 | 0.483         |
| C22 | 0.695         | C33 | 0.777         | C51 | 0.424         | C63 | 0.388         |

**Table 8**

Scheme dominance results under indicator C11.

(3) Calculate the dominance of the entire program

|    | A1     | A2     | A3     | A4    |
|----|--------|--------|--------|-------|
| A1 | 0      | −2.367 | −3.131 | 0.115 |
| A2 | 0.150  | 0      | −3.926 | 0.189 |
| A3 | 0.199  | 0.249  | 0      | 0.162 |
| A4 | −1.805 | −2.979 | −2.556 | 0     |

**Table 9**

Overall dominance of the scheme.

(4) Calculate the scheme sort value according to formula (15).

|    | A1      | A2     | A3      | A4      |
|----|---------|--------|---------|---------|
| A1 | 0       | −5.725 | −7.968  | −34.940 |
| A2 | −39.507 | 0      | −45.670 | −49.461 |
| A3 | −40.068 | −5.741 | 0       | −40.830 |
| A4 | −21.464 | −7.446 | −10.160 | 0       |

greater degree of discrimination. For instance, the difference between the best site A4 obtained by the TOPSIS method and the suboptimal solution A1 is 0.018, while the difference between the best site A4 obtained by the method in this paper and this optimal solution A1 is 0.1.

According to the above analysis, it is able to conclude that the method in this work considers the influence of the psychological factors of the decision makers, and has a good degree of discrimination.

## 7. Conclusion

In this paper, the low-carbon perspective is applied to the site selection of the BP project for the first time. Based on the literature review, a complete set of indicators is established, including 6 indicators and 20 sub-indicators. Then, the PLTSs are introduced to evaluate the ambiguity of the indicator, and the criteria weights are determined by the comprehensive integrated weighting method. Subsequently, a brand-new PLTS distance formula is defined to measure the deviation between evaluation information, based on it, an improved TODIM is applied to prioritize the alternatives. Eventually, the applicability of the framework in the practical application is illustrated by taking the Guangdong Province as a case, then the reliability of the framework is verified through sensitivity analysis and comparative analysis.

The advantages of the fuzzy multi-criteria decision-making method proposed in this paper can be summarized as the following aspects.

- The influence of low-carbon factors on the site selection of BP projects are considered in the criteria system, which expands the perspective of site selection and enriches the application of low-carbon theory in the field of energy.
- PLTSs are used to evaluate the uncertainty of evaluation information, which not only allow experts to express preferences in different linguistic terms, but also provide corresponding probabilistic information for each linguistic term, so as to better retain the original information.

**Table 10**

Project comprehensive ranking value.

| Scheme   | A1   | A2 | A3    | A4 |
|----------|------|----|-------|----|
| $S(A_p)$ | 0.90 | 0  | 0.502 | 1  |
| Sort     | 2    | 4  | 3     | 1  |

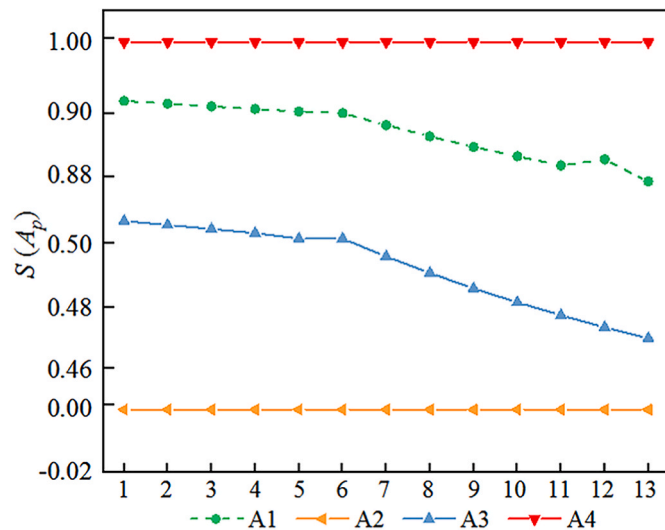


Fig. 4. Variation of scheme ranking value with  $\theta$ . 6.5.2 Change the weight of evaluation indicators.

Table 11  
Results of different decision methods.

| Alternative site | TOPSIS |      | VIKOR |      | Improved TODIM |      |
|------------------|--------|------|-------|------|----------------|------|
|                  | $S_i$  | Sort | $Q_i$ | Sort | $S(A_p)$       | Sort |
| A1               | 0.497  | 3    | 0.362 | 3    | 0.900          | 2    |
| A2               | 0.435  | 4    | 0.561 | 4    | 0              | 4    |
| A3               | 0.578  | 2    | 0.394 | 2    | 0.502          | 3    |
| A4               | 0.596  | 1    | 0.182 | 1    | 1              | 1    |

- (iii) The combination of ANP and entropy weight method enhances the rationality of weight determination for the final weights of all indicators.
- (iv) A new distance formula of PLTSs is defined. Based on this, an improved TODIM method is proposed to determine the optimal site, which is more in line with the real decision-making environment.

This paper also has some limitations which need further improvement. The criteria system established in this paper contains more qualitative indicators, the scoring of these qualitative indicators needs more in-depth understanding of the characteristics of BP projects. In the future, artificial neural networks can be used to simulate the psychological behavior of decision makers to ensure the accuracy of the model. Other multi-criteria decision-making methods can also be combined with the TODIM method to further improve the completeness of decision optimization and consider the combinatorial optimization problem of multiple projects.

#### Credit author statement

Shi-guanYang: Conceptualization, Investigation, Methodology, Writing – original draft. Jia-le Zhou: Software, Methodology, Writing – original draft. Zhuang Hu: Methodology, Investigation. Xin-yue Zhou: Conceptualization, Validation. Qi Cai: Methodology, Writing. Jin-heng Xie: Software, Writing. Yang-wen Wu: Supervision, Validation, Writing – review & editing. Qiang Lu: Conceptualization, Supervision, Funding acquisition, Writing – review & editing.

#### Declaration of competing interest

The authors declare that they have no known competing financial

interests or personal relationships that could have appeared to influence the work reported in this paper.

#### Data availability

Data will be made available on request.

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#### Appendix A. Supplementary data

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